

Design and Configuration of a Three-Building Campus Network

Technical Report

Mureșan Davide-Andrei

Supervisor: Prof. Dr. Eng. Adrian Peculea
Technical University of Cluj-Napoca

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1 Overview

1.1 Scope

A commercial institution occupies three buildings. Each building holds its own user population and must stay reachable if a single inter-building link fails. Four public services — web, file transfer, name resolution and mail — are published from a demilitarised zone, and a link to an Internet provider is established.

Three address blocks are given: 172.16.0.0/16 for the intranet, 210.1.1.64/27 for the DMZ, and 210.1.1.32/27 for the provider link.

1.2 What was built

Area	Implementation
Segmentation	Two VLANs per building — one for users, one for management — carried between switches over 802.1Q trunks.
Redundancy	Three switches per building cabled in a triangle, resolved by Spanning Tree; three routers in a full mesh, resolved by OSPF.
Addressing	DHCP served by each building router; infrastructure addresses static and excluded from the pools.
Access media	Cabled hosts and WPA2-protected wireless clients share the same user VLAN.
Routing	OSPF area 0 between the three building routers, with equal-cost multipath over the mesh.
Public services	HTTP, FTP, DNS and mail on public addresses in a dedicated DMZ, reachable both internally and from outside.
External link	Point-to-point connection to a provider router, with a simulated external network behind it.
Automation	Eight of the twelve device configurations generated from a single YAML inventory through Jinja2 templates.

1.3 Design decisions

Decision	Reason
Separate management VLAN	A compromised user host does not share a broadcast domain with the switch management plane.
802.1Q trunking	Without it, a VLAN cannot traverse a switch, and segmentation would be indistinguishable from separate physical networks.
/30 on point-to-point links	A link with two endpoints needs two usable addresses. A /24 would waste 252 per link.
Explicit spanning-tree root	Otherwise the Layer 2 topology is decided by MAC addresses, that is, by chance.
Dedicated DMZ switch	The DMZ is a separate Layer 2 domain with its own VLAN and managed interface, not servers attached to an internal switch.
Static addresses for servers	A service that changes address is a service nobody can find.
Configuration generated from an inventory	The addressing plan and the running configuration cannot drift apart.

1.4 Topology

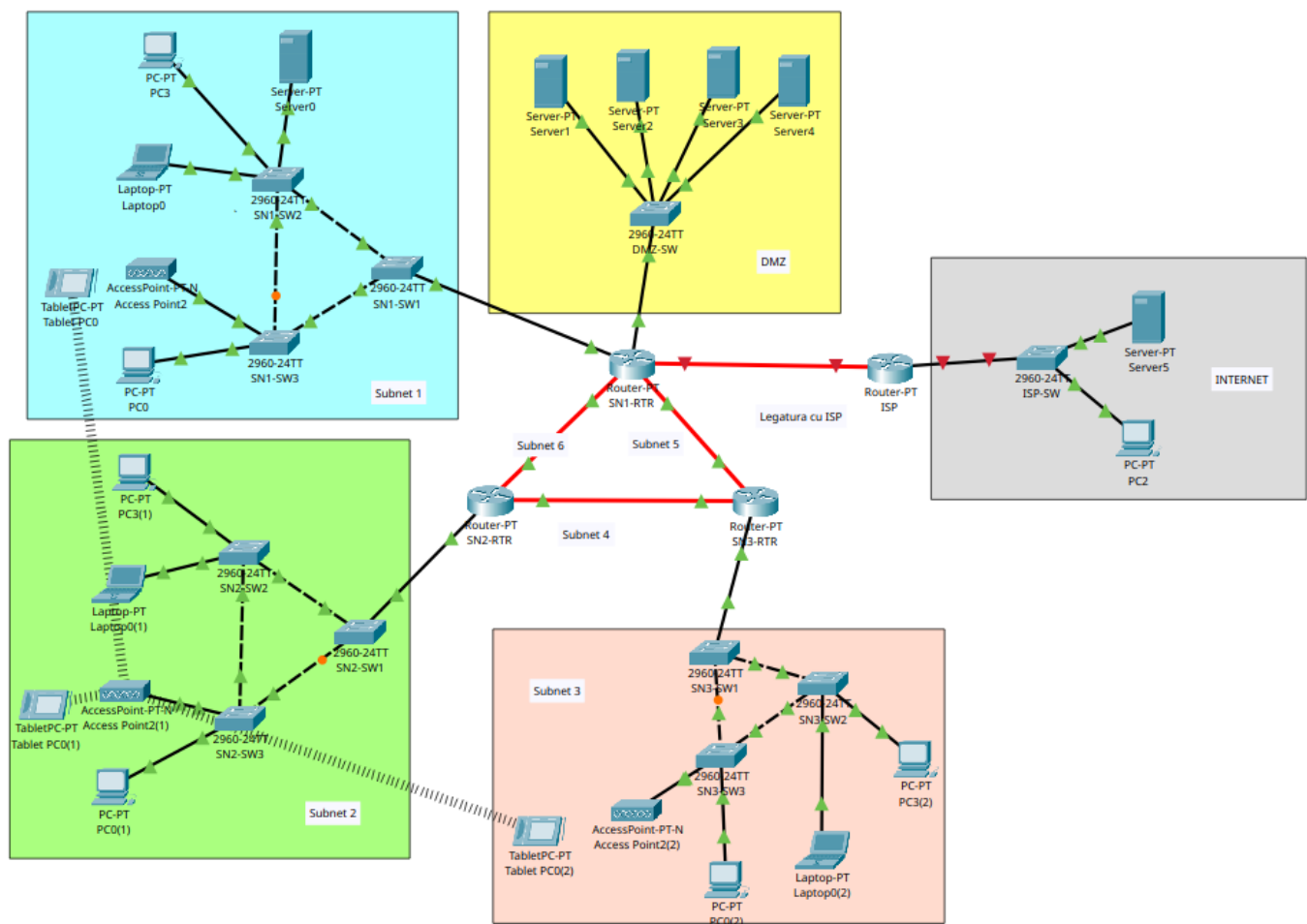


Figure 1: Network topology in Cisco Packet Tracer. Three user zones, a DMZ, and a simulated external network.

Table 1: Logical organisation

Zone	Router	Switches	User VLAN	Mgmt VLAN
Subnet 1	SN1-RTR	SN1-SW1/2/3	10	110
Subnet 2	SN2-RTR	SN2-SW1/2/3	20	120
Subnet 3	SN3-RTR	SN3-SW1/2/3	30	130
DMZ	on SN1-RTR	DMZ-SW	200	—
External	ISP	ISP-SW	—	—

SN1-RTR serves zone Subnet 1 and additionally holds the DMZ and the provider uplink.

1.5 Equipment

Four routers (Router-PT), eleven switches (Cisco 2960-24TT), three access points (AccessPoint-PT-N), six servers, and desktop, laptop and tablet clients.

2 Addressing Plan

2.1 Sizing

The number of usable hosts in a subnet with prefix p is $2^{(32-p)} - 2$. Two addresses are always unavailable: the network address, where all host bits are zero, and the broadcast address, where all host bits are one.

The specification requires at least 200 users per subnet:

Host bits	Prefix	Usable	
7	/25	126	insufficient
8	/24	254	smallest prefix that satisfies the requirement

2.2 Intranet

The /16 block yields 256 subnets of size /24 when eight bits are borrowed for the network portion. The third octet becomes the subnet number.

VLAN	Subnet	Purpose	Gateway
10	172.16.10.0/24	Subnet 1 users	172.16.10.1
20	172.16.20.0/24	Subnet 2 users	172.16.20.1
30	172.16.30.0/24	Subnet 3 users	172.16.30.1
110	172.16.110.0/24	Subnet 1 management	172.16.110.1
120	172.16.120.0/24	Subnet 2 management	172.16.120.1
130	172.16.130.0/24	Subnet 3 management	172.16.130.1
—	172.16.255.0/24	Point-to-point links	—

Two conventions apply. The VLAN identifier equals the third octet, so 172.16.20.55 identifies zone Subnet 2 without a lookup table. Infrastructure links occupy the last /24, leaving user subnets free to grow upward from 172.16.0.0.

Within each user subnet, .1-.9 are reserved for network equipment and excluded from DHCP; .10-.254 are allocated dynamically.

2.3 Point-to-point links

A router-to-router link has exactly two endpoints, so /30 is an exact fit: $2^2 - 2 = 2$ usable addresses. The block 172.16.255.0/24 divides into 64 such subnets, stepping by four.

Block	Network	Host A	Host B	Broadcast
172.16.255.0/30	.0	.1	.2	.3
172.16.255.4/30	.4	.5	.6	.7
172.16.255.8/30	.8	.9	.10	.11

The mask follows from the prefix: thirty network bits set to one and two host bits set to zero give 255.255.255.252, since $128 + 64 + 32 + 16 + 8 + 4 = 252$.

2.4 Public blocks

Both /27 blocks arrive already subnetted and their key addresses fixed.

Table 2: 210.1.1.32/27 — provider link

210.1.1.32	network address
210.1.1.33	ISP router
210.1.1.34	SN1-RTR, WAN interface
210.1.1.35 - .62	reserved for address translation
210.1.1.63	broadcast address

Table 3: 210.1.1.64/27 — DMZ

Address	Device	Name
210.1.1.65	SN1-RTR, DMZ interface	—
210.1.1.66	HTTP server	www.davide.ro
210.1.1.67	FTP server	ftp.davide.ro
210.1.1.68	DNS server	dns.davide.ro
210.1.1.69	Mail server	mail.davide.ro
210.1.1.70	DMZ-SW, management interface	—

Part I — Foundation

3 Naming and Topology Discovery

Devices arrived with factory default names: every router was **Router** and every switch **Switch**. The labels shown on the Packet Tracer workspace are independent of the configured hostname and do not appear in network protocols, so neighbour discovery was unusable:

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
Switch	Fas 1/0	138	S	2960	Fas 0/1
Switch	Fas 0/0	138	S	2960	Fas 0/4

Two different switches, one in Subnet 1 and one in the DMZ, reporting the same identifier.

3.1 Naming convention

Names follow the zones already present in the topology, so no translation is needed between the diagram and the command line.

SNx-RTR	zone router
SNx-SW1	switch connected to the router; spanning-tree root
SNx-SW2, SNx-SW3	access switches
DMZ-SW, ISP, ISP-SW	DMZ and external network

3.2 Configuration

On each of the fifteen devices:

```
enable
configure terminal
hostname SN2-RTR
end
```

Routers additionally required CDP to be enabled and their interfaces brought up. Unlike switch ports, router interfaces are administratively shut down from the factory:

```
configure terminal
cdp run
interface gigabitEthernet 6/0
no shutdown
interface gigabitEthernet 7/0
no shutdown
end
```

3.3 Discovery

With unique names in place, CDP became readable:

```
SN3-RTR#show cdp neighbors
```

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
SN2-RTR	Gig 6/0	145	R	PT1000	Gig 6/0
SN1-RTR	Gig 7/0	145	R	PT1000	Gig 7/0
SN3-SW1	Fas 0/0	145	S	2960	Fas 0/1

Queries from all four routers and all nine switches agree with one another, each link appearing from both ends with matching ports.

Table 4: Port map

Device	Interface	Neighbour	Remote	Purpose
SN1-RTR	Fa0/0	SN1-SW1	Fa0/4	LAN uplink
SN1-RTR	Fa1/0	DMZ-SW	Fa0/1	DMZ
SN1-RTR	Gi6/0	ISP	Gi6/0	WAN, fibre
SN1-RTR	Gi7/0	SN3-RTR	Gi7/0	transit
SN1-RTR	Gi8/0	SN2-RTR	Gi7/0	transit
SN2-RTR	Fa0/0	SN2-SW1	Fa0/3	LAN uplink
SN2-RTR	Gi6/0	SN3-RTR	Gi6/0	transit
SN3-RTR	Fa0/0	SN3-SW1	Fa0/1	LAN uplink
ISP	Fa0/0	ISP-SW	Fa0/1	external

Router-to-router links run on GigabitEthernet, access uplinks on FastEthernet — a fast core with slower edges, as in any campus network.

4 Routing Backbone

The three routers are cabled in a triangle. Any one link may fail without partitioning the network.

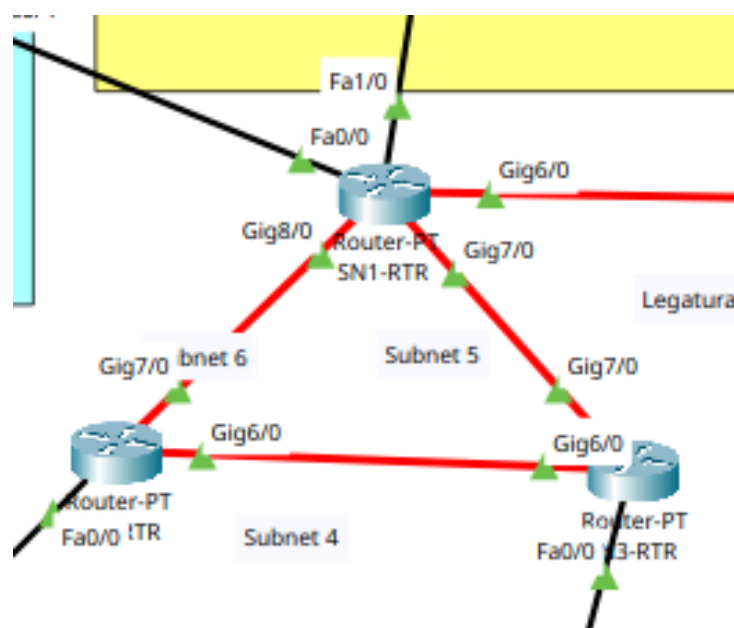


Figure 2: The three transit links between zone routers.

Subnet	Endpoint A	Endpoint B
172.16.255.0/30	SN1-RTR Gi8/0 → .1	SN2-RTR Gi7/0 → .2
172.16.255.4/30	SN2-RTR Gi6/0 → .5	SN3-RTR Gi6/0 → .6
172.16.255.8/30	SN1-RTR Gi7/0 → .9	SN3-RTR Gi7/0 → .10

Both endpoints of a link necessarily belong to the same /30; otherwise the link carries traffic while the routers fail to recognise each other as neighbours.

```
SN1-RTR(config)#interface gigabitEthernet 8/0
SN1-RTR(config-if)# description Transit to SN2-RTR
SN1-RTR(config-if)# ip address 172.16.255.1 255.255.255.252
SN1-RTR(config-if)# no shutdown
```

The `description` has no functional effect but records the purpose of the interface where it remains visible in `show running-config`.

4.1 Verification

```
SN1-RTR#show ip interface brief
Interface          IP-Address      OK? Method Status  Protocol
GigabitEthernet7/0 172.16.255.9    YES manual up      up
GigabitEthernet8/0 172.16.255.1    YES manual up      up
```

All three sides of the triangle respond:

Source	Destination	Success
SN1-RTR	172.16.255.2	100%
SN1-RTR	172.16.255.10	100%
SN2-RTR	172.16.255.6	100%

The first run of each test returns 80%: the initial packet is held while ARP resolves the destination hardware address. Subsequent runs return 100%.

4.2 Limits at this point

A ping from SN2-RTR to 172.16.255.10 fails:

```
SN2-RTR#ping 172.16.255.10
.....
Success rate is 0 percent (0/5)
```

This is correct behaviour. The router knows only its directly connected networks:

```
C    172.16.255.0/30 is directly connected, GigabitEthernet7/0
C    172.16.255.4/30 is directly connected, GigabitEthernet6/0
```

The address 172.16.255.10 belongs to the link between SN1-RTR and SN3-RTR, which SN2-RTR is not attached to. Without a routing protocol it has no way of learning that the network exists. This test is repeated once OSPF is running.

5 VLANs and Layer 2 Switching

Each zone holds three switches cabled in a triangle — a Layer 2 loop. Unlike IP packets, Ethernet frames carry no time-to-live field, so a single broadcast entering a loop circulates indefinitely, duplicating at every switch until the zone saturates.

Spanning Tree elects a root bridge and blocks the port that closes the loop, keeping it as a standby path.

5.1 Port roles

Access port	Belongs to one VLAN. End devices connect here and remain unaware that VLANs exist.
Trunk port	Carries several VLANs over one cable, each frame tagged with its VLAN identifier per IEEE 802.1Q.
SVI	A virtual interface giving the switch an address inside a VLAN. Needed for remote administration, not for forwarding.

5.2 Per-zone values

Zone	Users	Mgmt	SW1	SW2	SW3
Subnet 1	10	110	172.16.110.2	.3	.4
Subnet 2	20	120	172.16.120.2	.3	.4
Subnet 3	30	130	172.16.130.2	.3	.4

SW1 is root primary in every zone, SW2 is root secondary, SW3 keeps the default priority.

5.3 Configuration

Shown for SN2-SW1, which carries three trunks.

```

vlan 20
  name Users-SN2
exit
vlan 120
  name Mgmt-SN2
exit

interface fastEthernet 0/1
  description Trunk to SN2-SW2
  switchport mode trunk
  switchport trunk allowed vlan 20,120
exit

interface range fastEthernet 0/4-24
  switchport mode access
  switchport access vlan 20

```

```

exit

interface vlan 120
  ip address 172.16.120.2 255.255.255.0
  no shutdown
exit

ip default-gateway 172.16.120.1
spanning-tree vlan 20,120 root primary

```

Three details carry weight:

- A VLAN must exist on every switch its traffic crosses. Tagged frames for an unknown VLAN are discarded on arrival.
- `switchport trunk allowed vlan` restricts the link to the two VLANs in use. A trunk permits all 4094 by default, which both floods unnecessary traffic and opens a path for VLAN hopping.
- `switchport mode access` fixes the port role. Left unset, a port stays in dynamic mode and can be negotiated into a trunk by an attacking device.

Unused ports are configured as well. A port left in the default VLAN 1 is a port through which an unauthorised device reaches the network immediately.

5.4 Spanning-tree priorities

Switch	Command	Configured	Effective
SN2-SW1	root primary	$32768 - 8192 = 24576$	24596
SN2-SW2	root secondary	$32768 - 4096 = 28672$	28692
SN2-SW3	—	32768	32788

The effective value adds the VLAN identifier, since a separate spanning-tree instance runs per VLAN. SW1 is chosen as root because it holds the uplink to the router; placing the root elsewhere would make outbound traffic cross the zone twice.

5.5 Verification

```

SN2-SW1#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     20,120

```

```

SN2-SW3#show spanning-tree vlan 20
Root ID    Priority    24596
           Address    00D0.58B9.4CAC
           Cost        19

```

Port		4(FastEthernet0/4)			
Bridge ID	Priority	32788	(priority 32768 sys-id-ext 20)		
Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Desg	FWD	19	128.1	Shr
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/4	Root	FWD	19	128.4	P2p

The root is SN2-SW1; Fa0/4 is the root port, the lowest-cost path towards it; Fa0/2 is blocked. Exactly one port is blocked per zone, which is the expected result for a topology containing one loop.

5.6 Redundancy in effect

All three switches hold addresses in the management VLAN, so reachability between them exercises the whole Layer 2 configuration:

```
SN2-SW3#ping 172.16.120.3
!!!!
Success rate is 100 percent (5/5)
```

The result is more informative than it appears. 172.16.120.3 belongs to SN2-SW2, and the port connecting SN2-SW3 directly to it is blocked. The traffic therefore travels around the triangle, through SN2-SW1. The loop is eliminated while connectivity is preserved; should the active link fail, the blocked port is unblocked and traffic resumes over the standby path.

6 Inter-VLAN Routing

The zone router has one physical interface towards the zone but needs two addresses — a gateway for each VLAN. The interface is divided into logical sub-interfaces, one per VLAN. Frames arrive tagged over the trunk, and the router delivers each to the matching sub-interface.

```
interface fastEthernet 0/0
  no ip address
  no shutdown
exit

interface fastEthernet 0/0.20
  description Users VLAN 20
  encapsulation dot1Q 20
  ip address 172.16.20.1 255.255.255.0
exit

interface fastEthernet 0/0.120
  description Management VLAN 120
  encapsulation dot1Q 120
```

```
ip address 172.16.120.1 255.255.255.0
exit
```

The physical interface carries no address. The number after the dot is a local identifier, set equal to the VLAN for readability; the actual binding is made by `encapsulation dot1q`. Sub-interfaces inherit the state of the physical interface and need no `no shutdown`.

6.1 Verification

```
SN2-RTR#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet0/0.20	172.16.20.1	YES	manual	up	up
FastEthernet0/0.120	172.16.120.1	YES	manual	up	up

A host in the user VLAN reaching a switch in the management VLAN:

```
C:\>ping 172.16.120.3
Reply from 172.16.120.3: bytes=32 time=16ms TTL=254
Reply from 172.16.120.3: bytes=32 time<1ms TTL=254
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

The two hosts sit in networks separated at Layer 2, so the traffic must traverse the router. The TTL confirms it independently: each router decrements the field by one, and the reply arrives with 254 instead of 255. A ping to the gateway itself returns TTL=255, since no router is crossed.

6.2 Fault isolation

During verification, one test failed while its equivalents succeeded:

Source	Destination	Result
Host 172.16.20.50	gateway 172.16.20.1	success
SN2-RTR	switch 172.16.120.2	success
SN2-SW2	host 172.16.20.50	success
Host 172.16.20.50	switch 172.16.120.2	failure

The asymmetry locates the cause. The router shares a subnet with the switches and receives replies without the switch needing any routing knowledge. A host in a different subnet does not: the switch receives the request, but to answer it must forward the reply through its own default gateway. The switch in question was missing `ip default-gateway`, so requests arrived and replies never left.

A failed ping does not distinguish between a request that never arrived and a reply that never returned. Running the test from the opposite endpoint separates the two cases.

7 Dynamic Address Allocation

Each zone router serves DHCP for its own user VLAN. This works because the router is present inside the VLAN through its sub-interface and therefore receives the broadcast discovery messages directly. A server placed in another VLAN would have required `ip helper-address` on each sub-interface to convert those broadcasts into directed unicasts.

```
ip dhcp excluded-address 172.16.20.1 172.16.20.9

ip dhcp pool SN2-USERS
 network 172.16.20.0 255.255.255.0
 default-router 172.16.20.1
 dns-server 210.1.1.68
exit
```

`default-router` and `dns-server` do not configure the router; they are values transmitted to clients. The exclusion is a global command and must precede the pool — without it, allocation would begin at the gateway's own address.

The management VLAN is deliberately left out of DHCP. Infrastructure keeps static addresses, so a switch cannot lose its management address to an expired lease.

7.1 Verification

```
C:\>ipconfig /all
Physical Address.....: 0060.708B.8AE8
IPv4 Address.....: 172.16.20.10
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 172.16.20.1
DHCP Servers.....: 172.16.20.1
DNS Servers.....: 210.1.1.68
```

The allocated address is the first one after the excluded range, which demonstrates that the exclusion took effect. The DHCP server is the router itself, as required.

```
SN2-RTR#show ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
172.16.20.10    0060.708B.8AE8  --                Automatic
```

The hardware address in the lease table matches the one reported by the client.

7.2 A reporting quirk

`show ip dhcp pool` reports `Excluded addresses: 1` although nine addresses were excluded. Packet Tracer counts excluded *ranges* rather than individual addresses. The allocation beginning at .10 is the authoritative evidence.

8 Wireless Access

One access point per zone, connected to an access port in the user VLAN. Wireless clients therefore land in the same VLAN as cabled hosts and draw addresses from the same DHCP pool.

Zone	SSID	Authentication	Encryption
Subnet 1	SN1-WiFi	WPA2-PSK	AES
Subnet 2	SN2-WiFi	WPA2-PSK	AES
Subnet 3	SN3-WiFi	WPA2-PSK	AES

Access points are configured through the device interface rather than a command line. WPA2 with AES was selected over WPA with TKIP, which is deprecated.

8.1 Verification

A tablet associating and obtaining an address exercises the whole path at once: the WPA2 handshake, the access point bridging into the VLAN, the DHCP request reaching the router, and the reply returning.

```
C:\>ipconfig /all
IPv4 Address.....: 172.16.10.11
Default Gateway.....: 172.16.10.1
DNS Servers.....: 210.1.1.68
```

The address follows the one held by the cabled host in the same zone, confirming both media share a single pool. A ping from the tablet to that cabled host succeeds, confirming they share one VLAN rather than forming parallel networks.

9 OSPF

Without a routing protocol, each router knows only its directly connected networks. Reaching the other zones would require static routes on every router for every remote subnet — fifteen entries in this topology — and those routes would not adapt when a link fails, defeating the purpose of the triangular cabling.

OSPF is a link-state protocol: each router advertises its own links, all routers build the same map, and each computes shortest paths independently using Dijkstra's algorithm.

9.1 Configuration

```
router ospf 1
router-id 2.2.2.2
network 172.16.20.0 0.0.0.255 area 0
network 172.16.120.0 0.0.0.255 area 0
network 172.16.255.0 0.0.0.3 area 0
network 172.16.255.4 0.0.0.3 area 0
passive-interface fastEthernet 0/0.20
passive-interface fastEthernet 0/0.120
```

```
exit
```

<code>router-id</code>	Set explicitly. Left unset, the router derives it from an interface address, so its identity changes if that interface does.
<code>network ... area 0</code>	Selects which interfaces participate. The wildcard mask is the inverse of the subnet mask: 0.0.0.255 for a /24, 0.0.0.3 for a /30.
<code>passive-interface</code>	Suppresses Hello messages on user-facing sub-interfaces while still advertising their networks. No router lives there, and Hello packets on a user segment are both wasted traffic and free topology information for anyone listening.

Transit links were additionally declared point-to-point:

```
interface gigabitEthernet 6/0
ip ospf network point-to-point
```

OSPF treats an Ethernet segment as broadcast by default and elects a designated router. On a /30 holding exactly two routers this election is unnecessary and delays convergence.

9.2 Verification

```
SN3-RTR#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2 /0	1	FULL	00:00:32	172.16.255.5	GigabitEthernet6
1.1.1.1 /0	1	FULL	00:00:38	172.16.255.9	GigabitEthernet7

Two neighbours per router, both in FULL state — the databases are fully synchronised.

```
SN2-RTR#show ip route
```

```

172.16.0.0/16 is variably subnetted, 9 subnets, 2 masks
O    172.16.10.0/24 [110/2] via 172.16.255.1, GigabitEthernet7/0
C    172.16.20.0/24 is directly connected, FastEthernet0/0.20
O    172.16.30.0/24 [110/2] via 172.16.255.6, GigabitEthernet6/0
O    172.16.110.0/24 [110/2] via 172.16.255.1, GigabitEthernet7/0
C    172.16.120.0/24 is directly connected, FastEthernet0/0.120
O    172.16.130.0/24 [110/2] via 172.16.255.6, GigabitEthernet6/0
C    172.16.255.0/30 is directly connected, GigabitEthernet7/0
C    172.16.255.4/30 is directly connected, GigabitEthernet6/0
O    172.16.255.8/30 [110/2] via 172.16.255.1, GigabitEthernet7/0
                                [110/2] via 172.16.255.6, GigabitEthernet6/0
```

Routes marked 0 were learned through OSPF. The value [110/2] gives the administrative distance, 110 for OSPF, and the path cost.

The entry for 172.16.255.8/30 lists **two next hops of equal cost**. This is equal-cost multipath, a direct consequence of the triangular topology: the network is reachable through either neighbour at the same cost, and traffic is distributed across both.

The test that failed before OSPF was configured now succeeds, since SN2-RTR has learned the network it previously had no knowledge of.

Part II — Services

10 Demilitarised Zone

The four public servers occupy a separate Layer 2 domain with its own switch and VLAN, attached to SN1-RTR on a dedicated interface. Their addresses come from 210.1.1.64/27 and are public, so no translation is involved in reaching them.

VLAN 200 was chosen rather than 20, which is already in use as **Users-SN2** in another zone. The two switches share no trunk, so the identifiers could technically coexist, but reusing one across roles is a debugging trap.

```
! on SN1-RTR
interface fastEthernet 1/0
  description DMZ
  ip address 210.1.1.65 255.255.255.224
  no shutdown
exit

router ospf 1
  network 210.1.1.64 0.0.0.31 area 0
  passive-interface fastEthernet 1/0
exit
```

```
! on DMZ-SW
vlan 200
  name DMZ
exit

interface range fastEthernet 0/1-24
  switchport mode access
  switchport access vlan 200
exit

interface vlan 200
  ip address 210.1.1.70 255.255.255.224
  no shutdown
exit

ip default-gateway 210.1.1.65
```

No trunk is configured in the DMZ: a single VLAN has nothing to tag. No spanning tree concerns arise either, since there is one switch and no loop.

Advertising 210.1.1.64/27 into OSPF is what allows the other two zones to reach the servers. Without it, only SN1-RTR would know the network exists. The interface is passive because the DMZ holds servers, not routers.

Servers hold static addresses, with 210.1.1.65 as gateway and 210.1.1.68 as name server. The mask is 255.255.255.224; accepting the default /24 would make a server treat the entire 210.1.1.0 range as local and never use its gateway.

10.1 Verification

From the DMZ gateway:

Destination	Device	Success
210.1.1.66	HTTP server	80% then 100%
210.1.1.67	FTP server	80% then 100%
210.1.1.68	DNS server	80% then 100%
210.1.1.69	Mail server	80% then 100%
210.1.1.70	DMZ-SW	60% then 100%

From SN2-RTR, the DMZ prefix appears in the routing table as an OSPF route, and the servers respond — confirming that the advertisement reached the other zones.

11 Public Services

Server	Address	Service	Name
Server 1	210.1.1.66	HTTP, HTTPS	www.davide.ro
Server 2	210.1.1.67	FTP	ftp.davide.ro
Server 3	210.1.1.68	DNS	dns.davide.ro
Server 4	210.1.1.69	SMTP, POP3	mail.davide.ro

The DNS zone `davide.ro` holds four A records, one per service. Every host in the network received 210.1.1.68 as its name server through DHCP, so resolution works without any per-client configuration.

The FTP server holds an account with full read, write, delete, rename and list rights. The mail server runs SMTP for sending and POP3 for retrieval, with two mailboxes in the `mail.davide.ro` domain.

11.1 Verification

```
C:\>ping www.davide.ro
Pinging 210.1.1.66 with 32 bytes of data:
Reply from 210.1.1.66: bytes=32 time<1ms TTL=126
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

```
C:\>ping mail.davide.ro
Pinging 210.1.1.69 with 32 bytes of data:
Reply from 210.1.1.69: bytes=32 time<1ms TTL=126
```

The first line of each result shows the name resolved to an address, which is the evidence that the client queried the DNS server and received an answer. TTL=126 reflects two routed hops: the zone router, then SN1-RTR into the DMZ.

```
C:\>ftp ftp.davide.ro
Trying to connect...ftp.davide.ro
Connected to ftp.davide.ro
220- Welcome to PT Ftp server
Username:davide
331- Username ok, need password
Password:*****
230- Logged in
```

The web page is reachable by name from any host in the three zones.

12 External Connectivity

A point-to-point link over fibre connects SN1-RTR to a provider router, behind which a server and a workstation simulate an external network.

Device	Interface	Address
SN1-RTR	Gi6/0	210.1.1.34/27
ISP	Gi6/0	210.1.1.33/27
ISP	Fa0/0	100.0.0.1/8
External server	—	100.0.0.2/8
External host	—	100.0.0.3/8

```
! on SN1-RTR
interface gigabitEthernet 6/0
description WAN uplink to ISP
ip address 210.1.1.34 255.255.255.224
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 210.1.1.33
```

```
! on ISP
ip route 210.1.1.64 255.255.255.224 210.1.1.34
```

The WAN interface is deliberately excluded from OSPF. Routing information is not exchanged with a third party; the link is handled by a default route instead. 0.0.0.0 0.0.0.0 matches no bits and therefore matches everything, making it the route of last resort — used only when no more specific entry applies.

The provider needs the reverse direction. It is directly connected to 210.1.1.32/27 but has no knowledge of 210.1.1.64/27, so a static route points that prefix back through 210.1.1.34. This is the only static route on the provider side, and is justified: the provider does not run OSPF with the customer.

12.1 Verification

```
SN1-RTR#show ip route
Gateway of last resort is 210.1.1.33 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 210.1.1.33
```

The router reaches both external hosts. More significantly, the external server reaches the DMZ web server at 210.1.1.66, which exercises the static route on the provider and confirms that the public servers are visible from outside.

Hosts inside the three zones cannot reach the external network, and this is the expected result at this stage: they hold RFC 1918 private addresses, which are not routable beyond the site boundary. Address translation on the border router is the mechanism that resolves this.

Part III — Automation

13 Configuration Generation

Zone Subnet 2 was configured by hand, command by command, to establish a verified baseline and to understand each command before depending on it.

Repeating that across the remaining eight devices raises two problems. Transcription error: roughly twenty-five commands per device, each carrying VLAN identifiers, addresses and port numbers that differ between zones. And maintenance: a later change to the addressing plan would have to reach nine devices without missing one.

The remaining zones were therefore generated.

13.1 Structure

<code>inventory.yml</code>	The design: zones, VLAN identifiers, addresses, and the trunk ports of each device as discovered through CDP.
<code>templates/*.j2</code>	Jinja2 templates holding the IOS syntax, one per device role.
<code>render.py</code>	Loads the inventory, applies the templates, writes one configuration file per device.

Jinja2 is the same templating engine Ansible uses internally, so the templates would carry over to an Ansible workflow unchanged.

13.2 Derived values

Access ports are not listed in the inventory. The renderer computes them: any FastEthernet port not carrying a trunk becomes an access port in the user VLAN, and the resulting numbers are collapsed into the fewest contiguous `interface range` statements.

For a switch whose trunks occupy ports 1, 3 and 4:

```
interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 30
exit

interface range fastEthernet 0/5-24
  switchport mode access
  switchport access vlan 30
exit
```

Port 2 is emitted alone because it sits isolated between trunks; ports 5 to 24 are contiguous. Listing these by hand would mean recomputing the set whenever a cable moves.

13.3 Validation

Zone Subnet 2 remains in the inventory although it was built manually. Rendering it reproduces the hand-written configuration exactly, which is how the generator was verified before being pointed at untouched equipment.

The verification extends to the behaviour of the resulting network:

	Subnet 2 (manual)	Subnet 1 (generated)
Root bridge priority	24596	24586
Non-root switch priority	32788	32778
Blocked port	Fa0/2	Fa0/2
Root port	Fa0/4	Fa0/4

The priorities differ by ten, which is the difference between the two VLAN identifiers. The topologies converge identically.

13.4 Propagation and delivery

A change made once in the inventory reaches every affected device. Altering the name server address and re-rendering updates the DHCP pool on all three routers; none is edited individually and none can be forgotten.

Packet Tracer accepts no inbound SSH, since its devices exist only within the application and their addresses appear on no real interface of the host. Delivery is therefore performed by pasting each generated file into the device console. On production equipment the same files would be applied unchanged by Ansible, Netmiko or Nornir. The generation stage is unaffected by this limitation.

Twelve configurations are produced from 144 lines of inventory in under a tenth of a second.

14 Results

14.1 Requirement coverage

Requirement	Status	Where
Three user subnets, ≥ 200 hosts each	done	Sections 2, 5
Cabled and wireless access	done	Sections 5, 9
Redundancy in cabling and configuration	done	Sections 5, 10
DHCP served by the routers	done	Section 8
Public servers in a DMZ	done	Sections 11, 12
Domain name containing the student name	done	Section 12
OSPF routing	done	Section 10
ISP link, simulated external network	done	Section 13
WPA2 on wireless networks	done	Section 9
OSPF authentication	<i>planned</i>	—
Address translation towards the provider	<i>planned</i>	—
Privilege levels, password encryption, SSH-only management	<i>planned</i>	—
Two additional security measures	<i>planned</i>	—

14.2 Summary

The network is operational across all three zones. Each is segmented into user and management VLANs, protected against switching loops by a deterministically placed spanning tree, routed between VLANs by its own router, and addressed automatically. Wireless clients share the user VLAN with cabled hosts. The three zone routers exchange routes through OSPF over a redundant triangle that yields equal-cost paths, and every zone reaches the public servers in the DMZ by name.

Eight of the twelve device configurations were generated from a single inventory rather than typed, and the generated zones converge identically to the one built by hand.